C++ Complete Course

C++ → 1979 → Bjarne Stroustrope → extension of C

C++ ; Fast programs, more control over system resources and Memory management

High performance

major updates : 2011 C++ 11
2014 C++ 14
2017 C++ 17

Basic Structure

```
#include <iostream>        → Header file input/output (enhance Programming functionality)

(Blank lines ignore C++ files)
int main() {               → entry point (denotes main is a func^n)
    std::cout << "Hello world";   → func^n under standard dirspace
    return 0;              → Termination point
}
```

- Lowlevel : near to hardware
- Highlevel : Logically sound & away from hardware

Variables : containers to store data.

e.g int a = 3;
float c = 3.1;

single line comment //
multiline comment /* */
Select all + ctrl / → multiline comment

Program 1

```
# include <stdio.h>
using namespace std;
// this program was created by code with harry
/* This
 is a multiline comment */
int main() {
  cout << "Hello world";
return 0; }
```

Program 2 :

```
#include <stdio.h>
using namespace std;
int main() {
    int sum = 6;
    Cout << "Hello " << sum;
    return 0;
}
```

Output: Hello 6

VARIABLE SCOPE & DATA TYPES

- A variable is a container to hold data.
- A variable can be of various types. primarily we have these variable types in C++; i) int ii) Float iii) char iv) Double v) Boolean
- int sum = 34; meance sum is an integer variable which holds data 34 in memory.

Syntax

- Data-type variable-name = value;
  - Ex. int a = 4 , b = 6 ;

Based on scope, variable can be classified into two types:

1. Local variables
2. Global variables

VARIABLE SCOPE

- Scope of a variable is the region in code where the existence of variable is valid.
- Based on scope we have local and global variable in C++.
- Local variables: Local variables are declared inside the braces of any function and can be accessed only from there.
- Global variables: Global variables are declared outside any function and can be accessed from anywhere.
- Can global and local variable have same name in C++? Yes